

System Hardening Plan

Introduction

The BSF-Nutrifeed ecosystem is a digital platform supporting sustainable poultry feed production through data-driven monitoring and operations. As the system expands to include web applications, mobile interfaces, and networked devices, it becomes increasingly important to implement strong security measures. A well-defined system hardening plan helps protect sensitive data, maintain system integrity, and ensure reliable service delivery, particularly in rural environments with limited cybersecurity awareness.

System Hardening Measures

1. Device-Level Security

All devices should be regularly updated and patched to address known vulnerabilities. Strong authentication, including complex passwords and multi-factor authentication, should be enforced. Default settings and unnecessary services must be disabled to reduce attack surfaces. Endpoint protection tools such as antivirus software and host-based firewalls should also be implemented.

2. Network Protection Strategies

Network segmentation should be applied to isolate critical systems such as servers and databases. Firewalls and intrusion detection systems (IDS) should be used to monitor and control traffic. Secure communication protocols such as HTTPS and VPNs must be enforced to protect data in transit. Continuous monitoring and logging should be implemented to detect and respond to suspicious activities.

3. Secure Data Handling Practices

Sensitive data should be encrypted both at rest and in transit. Access control mechanisms must ensure that only authorized users can access critical information. Regular data backups should be maintained for recovery purposes. Additionally, secure coding practices and proper input validation should be implemented to prevent vulnerabilities such as SQL injection.

Conclusion

These measures will enhance the security and resilience of the BSF-Nutrifeed ecosystem against potential cyber threats.